Chenyu Gao

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Summary

Applied ML scientist and ECE PhD candidate at Vanderbilt University (expected 2026), with industry experience at TikTok (ByteDance) and insitro spanning multimodal LLM post-training for video understanding, reward modeling for video generation, and multimodal representation learning. Author of 40+ peer-reviewed publications, including 8 first/co-first-author papers, and inventor on a pending U.S. patent application.

Experience

Machine Learning Scientist Intern San Jose, CA
TikTok (ByteDance) *Jan 2026 – May 2026*

- **Latent-space reward modeling for video generation:** Developed a DiT-native reward model for spatiotemporal consistency and multi-object interaction fidelity, with a *Semantic Adapter* mapping VAE latents to semantic features distilled from a vision foundation model.
- Proposed early-step reward evaluation directly on VAE latents, exploiting early layout stabilization to reduce DiT denoising steps before scoring by 75% and eliminate VAE decoding.
- Achieved stronger agreement with human ratings than large VLM judges on target video qualities; used the reward model for GRPO-style RL post-training of video generation models.
- **LLM-driven video scoring:** Built an SFT pipeline to adapt a base LLM for multidimensional video-quality scoring, distilling rubric-guided rationales into concise outputs for higher-QPS serving.
- Built a pipeline supplying LLM-derived video-quality features to production recommendations, contributing to a 4.15% relative increase in feature penetration and platform-wide lifts in publish frequency (+0.03%) and search engagement (+0.019%).

Data Science and Machine Learning Intern South San Francisco, CA
insitro *Jun 2025 – Aug 2025*

- **Image–genetics self-supervised learning:** Designed a multimodal SSL framework for target discovery, combining masked image modeling with SNP-guided contrastive alignment from whole-genome sequencing data.
- **LLM-assisted genomic feature selection:** Built a human-in-the-loop LLM/ontology pipeline that reduced 1B+ SNPs from whole-genome sequencing to a curated panel of hundreds, using published genetic associations to prioritize image-observable phenotypes and make multimodal SSL computationally tractable.
- Converted learned image embeddings into disease-relevant quantitative phenotypes via regression and clustering, enabling GWAS in imaging cohorts to recover candidate genetic associations and surface target-discovery hypotheses.

Graduate Research Assistant Nashville, TN
Vanderbilt University *Jul 2022 – Present*

- **Large-scale ML infrastructure:** Built containerized processing, harmonization, and QC workflows for 100,000+ MRI scans from 40+ datasets across local and Slurm/HPC environments, enabling reproducible multi-cohort ML studies.
- **Generative AI privacy and safety:** Led MRI de-identification studies evaluating the privacy–utility tradeoff of defacing methods; built a cascaded diffusion-model red-team pipeline that reconstructs high-resolution 3D facial geometry from defaced head MRI, exposing re-identification risk and motivating privacy safeguards for medical data sharing.
- **Multimodal predictive modeling:** Developed brain-age models using diffusion and structural MRI from 13,398 participants across 12 datasets to support early detection of neurodegenerative disease; this work resulted in a U.S. patent application.
- **Generative modeling:** Co-developed conditional GAN models to extend incomplete MRI fields of view, rescuing otherwise unusable scans and increasing effective dataset size by 10%; also co-developed a diffusion model for joint modeling of imaging and clinical metadata.

Graduate Research Assistant Baltimore, MD
Johns Hopkins University *Dec 2020 – May 2022*

- **Continual learning:** Extended a lifelong learning algorithm from vision to speech and validated omnidirectional knowledge transfer on a spoken-digit benchmark, contributing to an open-source library (link) and a *TPAMI* paper.
- **Medical image analysis:** Benchmarked deep learning and computer vision methods for MRI analysis, establishing baselines for subsequent work on MRI privacy, defacing, and data utility evaluation.

Education

Vanderbilt University — Nashville, TN
Doctor of Philosophy in Electrical & Computer Engineering — *Jul 2022 – Dec 2026 (expected)*

Johns Hopkins University — Baltimore, MD
Master of Science in Biomedical Engineering — *Aug 2020 – May 2022*

Sun Yat-sen University — Guangzhou, China
Bachelor of Science in Biomedical Engineering — *Aug 2016 – Jun 2020*

Selected Publications & Patent

Full list: 40+ peer-reviewed publications, including 8 first/co-first-author works; see Google Scholar.

S1. **C Gao***, K Xu*, et al. "Pitfalls of defacing whole-head MRI: re-identification risk with diffusion models and compromised research potential." *Computers in Biology and Medicine*. 2025. **[Diffusion-model privacy red-team; generative AI safety]**

S2. **C Gao**, et al. "Brain age identification from diffusion MRI synergistically predicts neurodegenerative disease." *Imaging Neuroscience*. 2025. **[Multimodal predictive modeling; related U.S. patent application]**

S3. **Patent: C Gao**, BA Landman, ME Kim. "System and Method of Brain Age Identification for Predicting Neuro-Degenerative Disease." U.S. Patent Application No. 19/345,706. **[Lead inventor; applied predictive ML]**

S4. **C Gao**, et al. "Field-of-view extension for brain diffusion MRI via deep generative models." *Journal of Medical Imaging*. 2024. **[Conditional generative modeling]**

S5. Y Liu, **C Gao**, et al. "MetaVoxel: Joint Diffusion Modeling of Imaging and Clinical Metadata." *Medical Imaging with Deep Learning*. 2026. **[Multimodal diffusion modeling]**

S6. JT Vogelstein, J Dey, …, **C Gao**, et al. "Simple Lifelong Learning Machines." *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2025. **[Continual learning]**

S7. ME Kim, **C Gao**, et al. "White matter micro- and macrostructure brain charts for the human lifespan." *Nature*. 2026. **[Population-scale ML infrastructure; normative brain-chart modeling]**